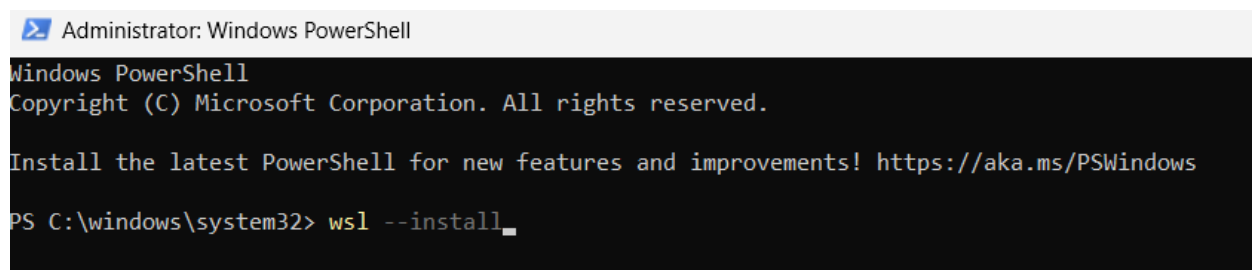
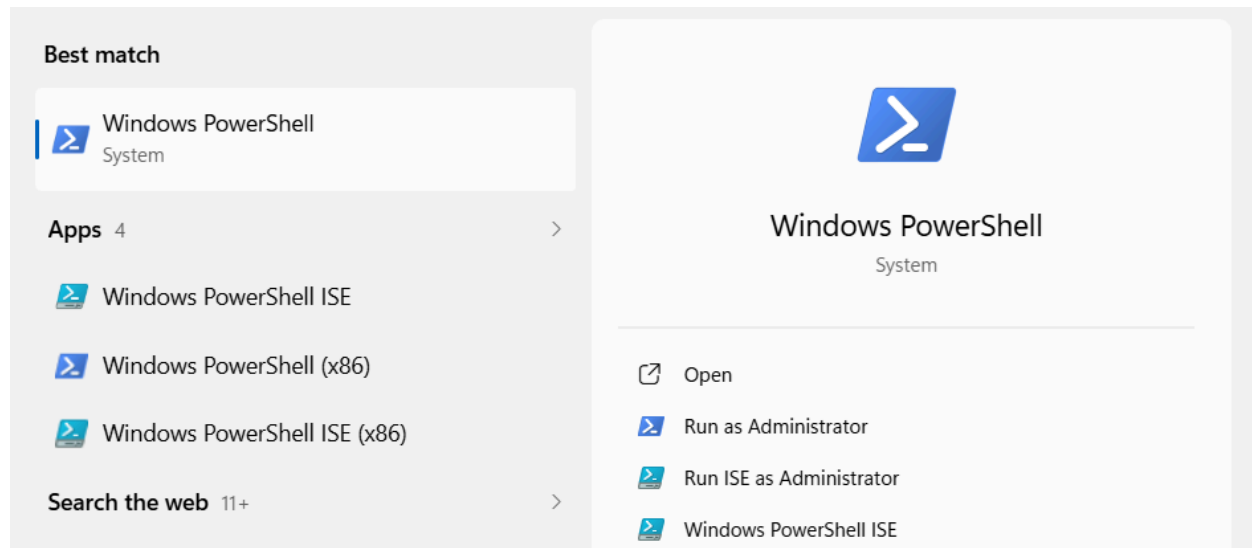


Step 1: Open PowerShell as Administrator

1. Press **Windows**.
2. Search for **PowerShell**.
3. Click on **Run as Administrator**.



Step 3: Restart Your Computer

After installation, restart your PC when prompted.

Step 4: Complete Ubuntu Setup

After restarting:

1. Ubuntu opens automatically (or search for **Ubuntu** in the Start menu).
2. Wait for the installation to finish.
3. Create a Linux username.

Step 5: Create a Password

You'll then be prompted to create a password.

New password: ****

Retype new password:****

Step 6: Update Ubuntu

```
abhis@MIKHAIL-37: ~  
abhis@MIKHAIL-37:~$ sudo apt update  
[sudo: authenticate] Password:  
Get:1 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]  
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease  
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]  
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]  
Get:5 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]  
Hit:6 http://archive.ubuntu.com/ubuntu resolute-backports InRelease  
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [492 kB]  
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]  
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [243 kB]  
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [184 kB]  
Fetched 1382 kB in 3s (477 kB/s)  
115 packages can be upgraded. Run 'apt list --upgradable' to see them.  
abhis@MIKHAIL-37:~$ |
```

Step 7: Install GCC (C Compiler)

```
abhis@MIKHAIL-37:~$ sudo apt install build-essential  
build-essential is already the newest version (12.12ubuntu2.26.04.2).  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 115  
abhis@MIKHAIL-37:~$ |
```

Step 8: Verify the Installation

```
abhis@MIKHAIL-37:~$ gcc --version  
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0  
Copyright (C) 2025 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Step 9: Create Your C Program

Example: nano process.c

Step 10: Compile the Program

```

abhis@MIKHAIL-37:~$ nano process.c
abhis@MIKHAIL-37:~$ gcc process.c -o process
./process
Enter a Linux command: pwd

Parent Process
Parent PID : 986
Child PID  : 987

Child Process
Child PID : 987
Parent PID: 986
/home/abhis

Child process completed.
abhis@MIKHAIL-37:~$ |

```

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kumarabhishek-crypto Delete PRACTICAL/README.md		ba84f3d · last week	🕒 15 Commits
 PRACTICAL	Delete PRACTICAL/README.md	last week	
 SKILL	Create README.md	2 weeks ago	

1. Execute `fork()` & Analyze Process Abstraction

```

abhis@MIKHAIL-37:~$ nano process.c
abhis@MIKHAIL-37:~$ gcc process.c -o process
abhis@MIKHAIL-37:~$ ./process
Parent [PID: 1201 | Child PID: 1202] waiting...
Child [PID: 1202 | PPID: 1201] running...
total 24
drwxr-xr-x 2 abhis abhis 4096 Aug 23 13:35 OS
-rwxr-xr-x 1 abhis abhis 16248 Aug 24 05:27 process
-rw-r--r-- 1 abhis abhis 667 Aug 24 05:27 process.c
Child process completed.
abhis@MIKHAIL-37:~$ |

```

2. Inspect Process Tree & Parent-Child Relationships

```

abhis@MIKHAIL-37:~$ sudo apt update && sudo apt install psmisc
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
115 packages can be upgraded. Run 'apt list --upgradable' to see them.
psmisc is already the newest version (23.7-2ubuntu2).
psmisc set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 115
abhis@MIKHAIL-37:~$ ./process
Parent [PID: 1409 | Child PID: 1410] waiting...
Child [PID: 1410 | PPID: 1409] running...
total 24
drwxr-xr-x 2 abhis abhis 4096 Aug 23 13:35 OS
-rwxr-xr-x 1 abhis abhis 16248 Aug 24 05:27 process
-rw-r--r-- 1 abhis abhis 667 Aug 24 05:27 process.c
Child process completed.

```

3. Practice System Call Tracing

```
abhis@MIKHAIL-37:~$ sudo apt install strace
Installing:
  strace

Installing dependencies:
  libunwind8

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 115
  Download size: 737 kB
  Space needed: 2660 kB / 1024 GB available

Continue? [Y/n] Y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 strace amd64 6.19+ds-0ubuntu5 [678 kB]
Fetched 737 kB in 1s (1076 kB/s)
Selecting previously unselected package libunwind8:amd64.
(Reading database ... 42277 files and directories currently installed.)
Preparing to unpack .../libunwind8_1.8.3-0ubuntu1_amd64.deb ...
Unpacking libunwind8:amd64 (1.8.3-0ubuntu1) ...
Selecting previously unselected package strace.
Preparing to unpack .../strace_6.19+ds-0ubuntu5_amd64.deb ...
Unpacking strace (6.19+ds-0ubuntu5) ...
Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
Setting up strace (6.19+ds-0ubuntu5) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.3) ...
```